

Unit 3 - AM

MC - Questions Answered
on D2L

MC	Written	Total
/14	/21	/35

Knowledge/Application

- Solve the inequality, $3 \leq \frac{6}{x}$
- Given the function, $g(x) = \frac{x+1}{x^2+4x-5}$, complete the chart and graph the function.

Application

- Solve the inequality $\frac{(2x-1)(x+3)}{(x-2)(x-5)} \geq 0$ using an interval table.
- The population of tadpoles in a pond could be modeled by the function $f(t) = \frac{40}{t+1}$ where t is given in days. The population of frogs in the same pond can be modelled by the function $g(t) = \frac{45t}{t^2+8t+7}$. Determine when there will be more frogs than tadpoles in the pond.

Communication

- Why do graphs of reciprocals of linear functions always have vertical asymptotes, but the graphs of reciprocal quadratic functions sometimes do not? Provide sketches of each situation to support your answer.
- Mr. Azar claims that if you were to graph $f(x) = \frac{x^2-49}{x+7}$, you would get an oblique asymptote since the degree in the numerator is greater than the degree in the denominator. Is he correct? If not justify.

Thinking

- Graph the function $f(x) = \frac{x^3+x^2-5x+3}{x^2+5x+6}$. State the key features and label them on the graph.
- If the function $f(x) = \frac{ax+b}{(x-1)(x-4)}$, has a horizontal asymptote at $y = 2$ and a y-int of $\frac{1}{4}$, determine the value of $a \times b$?